

Barkley Heights Retaining Wall Site Inspection Report

Draft date: 2026-08-17

Prepared for: Admiral Insurance / Counsel

Prepared by: David Salter, DRS Engineering

Project: Barkley Heights Development, Bellingham, Washington

Inspection date: 2026-07-23

Draft Status

This is a working draft prepared from dictated field notes, photo captions, preliminary document-review notes, and the existing evidence log. It should not be issued as a final expert report until Dave reviews and confirms the factual record, opinions, assumptions, and any required expert disclosures.

Items still requiring confirmation are listed in Section 10.

1. Assignment and Purpose

DRS Engineering was retained in connection with reported concerns involving the retaining/shoring wall system at the Barkley Heights development in Bellingham, Washington. The site visit was conducted to observe the retaining wall area, adjacent uphill property conditions, apparent wall alignment, and available site conditions relevant to reported movement or distress.

The wall at issue is understood from the received background materials to be associated with Lots 11, 12, and 13, downslope of the Bearden property. The visit focused on visual observations and limited field measurements. This report does not present a full independent design check unless specifically supplemented with calculation review, design drawings, geotechnical reports, construction records, and survey data.

2. Materials Considered

The following materials were available in the current project file and were used to prepare this draft:

- Dave Salter's dictated field notes from the 2026-07-23 Barkley Heights site visit.
- Site photographs identified in the evidence log as P001 through P018.
- Dave's captions and post-photo comments for selected photographs.
- Dave's field measurement/estimate notes identified as M001 and M002.
- Dave's summary of his conversation with David Ebenal, identified as C001.
- A brief history/background document received from Erin, identified as D001.
- DEI/Dibble Engineers' 2026-07-09 structural review, identified as D002.

- Preliminary internal notes reviewing selected DEI calculation issues.

The current local project folder contains descriptions and media references for the site photographs, but not the full original image files. The photo appendix in this draft therefore identifies each image by evidence number and description, pending insertion of the actual images.

3. Site Visit Chronology

Dave arrived at the Barkley Heights site at approximately 6:05 p.m. PT on 2026-07-23. He noted that David Ebenal was expected to meet him shortly and that subsequent comments and photographs should be treated as job-site observations.

General arrival photographs were taken at approximately 6:06 to 6:08 p.m. PT. Those photographs documented the retaining/shoring wall area, adjacent partially constructed residential foundations, wood lagging between vertical steel members, lower concrete wall/foundation elements, gravel backfill, black waterproofing or drainage sheeting, white pipes, covered construction materials, and existing uphill residences or vegetation.

Dave reported an approximately 15-minute conversation with David Ebenal between about 6:15 and 6:30 p.m. PT. Dave described Ebenal as reluctant to provide much detail concerning the sequence of events. According to Dave's summary, Ebenal generally described the sequence as "we built this wall and now we're going to add something to it." When asked about initial slopes, Ebenal reportedly acknowledged that there had been a slope. Dave asked whether there were problems with the neighbor's property during slope excavation and before the wall was built; Dave reported that Ebenal did not provide a detailed sequence or much information.

After that conversation, Dave continued photographing and documenting conditions along the wall and uphill property area. He left the site at approximately 7:00 p.m. PT.

4. Observed Site Conditions

4.1 General Wall and Construction Area

The arrival photographs document a retaining/shoring wall system adjacent to partially constructed residential work. Visible conditions include upper wood lagging spanning between vertical steel members, a lower concrete wall or wall/beam element, adjacent foundation construction, gravel fill or backfill near the wall, black waterproofing or drainage membrane/sheeting, and white drainage or plumbing pipes.

Photographs P001 through P005 and P007 provide general context for the wall alignment and surrounding construction. Those photographs should be treated as visual documentation unless tied to specific measurements or observed distress.

4.2 Apparent Wall Waviness and Pile Alignment

Dave identified photograph P006 as showing the apparent "wigglyness" of the wall. Photograph P007 provides additional close along-wall context, with the wall line receding into the distance. Photograph P013 was identified by Dave as showing misalignment of piles at the south end.

Dave later reported measuring soldier pile verticality approximately 3 feet above the connection to the concrete wall/wall-beam area. According to the field note, the measured piles leaned toward the uphill side by approximately 0.6 to 1.4 degrees.

Dave's field interpretation was that those measurements did not indicate clear evidence of outward wall movement due to backfill or excessive deflection of the soldier piles during backfilling. He also noted that the wall irregularity could be the result of poor initial alignment of soldier piles when they were placed in drilled shafts.

That interpretation should be confirmed against any available survey data, as-built records, pile-by-pile measurements, and construction sequence records. Without that additional information, the photographs and measurements support the observation that the wall line appears irregular, but they do not by themselves establish the timing or cause of that irregularity.

4.3 North End Conditions

Photograph P008 was identified by Dave as showing the proximity of the retaining/shoring wall to the house at the north end. The photo description indicates a narrow space between the residence and the wall, with the residence exterior/eaves and patio slab on one side, and the retaining/shoring wall on the other.

Photograph P014 was identified by Dave as showing double piles at the north end. The photograph documents paired steel pile elements and related top-of-wall conditions. No structural adequacy conclusion should be drawn from that photograph alone without comparison to the permitted design, retrofit drawings, and as-built records.

4.4 South End and Uphill Property Conditions

Dave identified several photographs as documenting conditions at or near the south end and the uphill property above the wall:

- P009 : settlement of the porch in the property above the wall at the south end.
- P011 : Eco blocks at the top of the slope at the south end.
- P013 : misalignment of piles at the south end.
- P015 : the house above the property propped up.
- P016 : a possible head scarp or head-scarp-like feature of a failure beyond the property line, with Eco blocks missing.
- P017 : porch column foundations undermined.
- P018 : original back cut.

Those observations indicate that the uphill property and top-of-slope area had visible support, slope, and foundation-related conditions warranting further review. In particular, the reported porch settlement, propping, possible head scarp, missing Eco blocks, and undermined porch column foundation should be correlated with prior photographs, construction records, repair history, city or neighbor reports, and survey data before any final opinion is offered regarding timing, causation, or structural significance.

4.5 Back-Cut Slope

Dave reported that the back-cut slope at the area where black plastic or black erosion-control material covered part of the upper exposed slope appeared to indicate an original slope angle of approximately 55 to 70 degrees.

That field estimate appears to be related to the original back-cut condition shown in P018 and to the nearby uphill property/slope conditions shown in P015, P016, and P017. Before using the angle estimate in any calculation, Dave should confirm the exact measurement location, method, instrument, and whether the angle was measured from horizontal.

5. Preliminary Document Context

The received background document D001 describes the Barkley Heights development on Summit Road in Bellingham, Washington, with issues involving Lots 11, 12, and 13 adjacent to the Bearden property. The document identifies Barkley Heights II, LLC, Bradley Engineering / David Bradley, Summit Construction Group, uphill neighbors Amy and Brad Bearden, and DRS Engineering's retained expert role for Admiral Insurance.

The background document's timeline indicates that exterior site work began in summer 2023; excavation on Lots 11 through 13 began in early 2025; neighbor complaints and city notice reportedly occurred in summer 2025; Barkley engaged David Bradley in August 2025; the Beardens alleged fissures and partial wall collapse or ecology-block movement in September 2025; new wall construction was reportedly completed in October 2025; a Certerra inspection report was dated 2025-10-27; the Bearden suit was filed on 2025-11-04; Barkley separately sued Bradley on 2026-06-17; and retrofit construction was reportedly scheduled to begin on 2026-07-27.

Those chronology items should be treated as background allegations or reported history unless independently verified through project records, pleadings, city documents, photographs, survey data, inspection reports, or witness testimony.

DEI/Dibble Engineers' 2026-07-09 structural review D002 describes a continuous soldier pile shoring wall across Lots 11, 12, and 13, with documentation and permitting separated by lot. DEI states that Lot 11 retained approximately 18 feet 6 inches of soil and Lots 12 and 13 retained approximately 14 feet 6 inches. DEI identifies principal elements as treated wood lagging, a lower concrete wall/lagging section, wide flange soldier piles embedded 10 feet, an 8-inch concrete footing, Lot 11 counterfort walls with diagonal HSS braces and threaded rods, and concrete fin walls at the north and south ends.

DEI's review raises concerns regarding soil pressure assumptions, seismic surcharge, static equilibrium, load path, component overstress, pile embedment, concrete wall/footing capacity, global stability, and brace force transformation. A preliminary internal review of DEI's visible arithmetic did not identify an obvious arithmetic error undermining DEI's main criticisms. Two apparent DEI presentation issues were identified: a Lot 12 denominator typo in a percent-unaccounted calculation, where the final percentage appears correct, and wording stating that factors of safety "exceed" allowable values where DEI likely meant that they did not meet required values.

This draft does not adopt DEI's opinions as DRS opinions. DEI's findings should be reviewed against the complete Bradley drawings and calculations, geotechnical recommendations, construction records, and DEI's native calculation inputs before final opinions are issued.

6. Preliminary Opinions and Observations

Based on the current field notes and photo descriptions, the following preliminary observations appear supportable for draft discussion:

1. The retaining/shoring wall line appeared visibly irregular in the field, and Dave documented apparent wall waviness and pile misalignment in photographs.
1. Dave's limited field verticality measurements, reportedly taken approximately 3 feet above the concrete wall/wall-beam connection area, indicated that measured soldier piles leaned toward the uphill side by approximately 0.6 to 1.4 degrees.
1. Those limited verticality measurements do not appear to show clear evidence of outward wall movement caused by backfill pressure or excessive soldier pile deflection during backfilling.
1. The observed wall irregularity could be consistent with poor initial soldier pile alignment during drilled-shaft placement, but that explanation should be verified against survey/as-built records, pile installation records, construction photographs, and any measurements made before and after backfilling.
1. Conditions at the uphill property, including reported porch settlement, apparent propping, possible head scarp/failure feature, missing Eco blocks, and undermined porch column support, warrant further review because they may be important to the chronology and causation issues.
1. The reported original back-cut slope angle of approximately 55 to 70 degrees is potentially significant, but it must be confirmed before being used for engineering calculations or stability opinions.
1. The conversation with David Ebenal, as summarized by Dave, did not provide a detailed construction or distress sequence. That absence of a detailed sequence increases the importance of obtaining contemporaneous records, photographs, inspection reports, city correspondence, survey data, and witness testimony.

7. Limitations

This draft is based on currently available dictated notes, photo descriptions, and preliminary document summaries. The current project folder does not contain the original full-size site photograph files. The report should be updated once those images are exported and inserted.

No final structural adequacy opinion is offered in this draft. A final opinion may require:

- Original design drawings and calculations by Bradley Engineering.
- Geotechnical reports and recommendations, including GeoTest and GeoEngineers materials referenced by DEI.
- DEI retrofit drawings, calculations, and native calculation inputs.
- Certerra inspection reports, concrete/steel/density reports, and other special inspection records.
- Construction sequence records, RFI/submittal records, and dated construction photographs.
- Survey data or measured wall/pile alignment records.
- Prior photographs and records concerning the Bearden property, porch/deck supports, Eco blocks, and alleged slope or wall movement.

- City notices, complaints, permit records, and related correspondence.
- Clarified testimony or interview notes from David Ebenal and other project participants.

8. Recommended Next Steps

1. Export and preserve the original site photographs corresponding to P001 through P018 .
1. Insert selected photographs into the report body, with the remaining photos placed in a photo appendix.
1. Confirm the exact client, case caption, report audience, and required expert report format.
1. Confirm the inspection scope and identify each wall segment by lot, orientation, and station or photo reference.
1. Confirm Dave's measurement tool, sign convention, pile-by-pile verticality values, and measurement locations.
1. Obtain or verify the Bradley drawings/calculations, geotechnical reports, construction records, survey/as-built data, Certerra reports, DEI retrofit package, and available prior photographs.
1. Decide whether the report should include a full independent calculation review or remain focused on site-observation and chronology issues.

9. Draft Evidence and Photo Appendix

ID	Description / Draft Use
P001	General arrival overview of retaining/shoring wall area and adjacent partially constructed foundation.
P002	Overview from lower grade showing retaining/shoring wall, uphill residence/deck, foreground framing/foundation work, construction materials, vehicles, and adjacent residences.
P003	Centered overview of wall below uphill residence/deck, including foreground construction, blue dumpster, ladder, covered materials, and exposed soil/rock.
P004	Overview/detail of wall and stepped foundation/backfill area, including black waterproofing/drainage membrane or sheeting, gravel backfill, white pipes, and exposed soil/rock.
P005	Close along-wall view showing wood lagging between vertical steel members, lower concrete wall, backfill/gravel path, adjacent foundation/building area, and temporary guardrail.
P006	Dave-identified photo showing apparent wall "wigglyness."
P007	Close along-wall view documenting wall alignment context.
P008	Dave-identified photo showing proximity of the wall to the house at the north end.
P009	Dave-identified photo showing settlement of the porch in the property above the wall at the south end.
P010	Along-grade photo of existing uphill concrete retaining/site wall, gravel path, black fabric/sheeting, and downhill construction.

ID	Description / Draft Use
P011	Dave-identified photo of Eco blocks at the top of the slope at the south end.
P012	Dave-identified photo of gravel fill above the wall.
P013	Dave-identified photo of pile misalignment at the south end.
P014	Dave-identified photo of double piles at the north end.
P015	Dave-identified photo showing the house above the property propped up.
P016	Dave-identified possible head scarp/failure area beyond property line, with Eco blocks missing; timing unknown.
P017	Dave-identified porch column foundation undermining.
P018	Dave-identified original back-cut condition.
M001	Dave's estimate that the back-cut slope where black plastic/erosion-control material was present may have been approximately 55 to 70 degrees.
M002	Dave's measurement note that soldier piles approximately 3 feet above the concrete wall/wall-beam connection leaned toward the uphill side by approximately 0.6 to 1.4 degrees.
C001	Dave's summary of an approximately 15-minute conversation with David Ebenal.

10. Open Items Before Finalizing

- Confirm client/owner and formal report addressee.
- Confirm the correct case name, claim number, and litigation caption if required.
- Confirm whether the project should be referred to as Barkley Heights, Barclay Heights, or another formal name from the pleadings/contracts.
- Confirm Dave's exact retained role and opinion scope.
- Confirm whether all of Lots 11, 12, and 13 are within the report scope.
- Confirm weather and general site conditions during the inspection.
- Confirm the measurement method and source data for M001 and M002.
- Confirm whether any of Dave's preliminary observations should be elevated into final opinions.
- Confirm whether DEI's calculation review should be incorporated into the report or kept as separate work product.
- Export and insert the original full-size site photographs.